

SymPy Bug Card

Bug 7: ODE solver loses signs in $f' = |f|$

Task. Solve the first-order ODE

$$f'(x) = \sqrt{f(x)^2}.$$

SymPy's answer.

$$f(x) = C_1 e^{-x} \quad \text{or} \quad f(x) = C_1 e^x, \quad C_1 \in \mathbb{R}.$$

Correct answer.

$$f(x) = 0, \quad f(x) = C e^x \ (C > 0), \quad f(x) = C e^{-x} \ (C < 0).$$

Explanation. For real-valued solutions, the equation is $f' = |f|$. A nonzero solution cannot change sign, so a negative solution satisfies $f' = -f$ and a positive solution satisfies $f' = f$.

Likely root cause. In `sympy/solvers/ode/ode.py`, `odesimp` appears to force a branch-sensitive implicit separable solution through `solve(..., force=True)` and then returns explicit branches without the required sign condition. This is a new instance of a known failure mode.

Artifact (GitHub). [bug-report/bug-007-ode-solver-loses-signs-in-f-abs-f/](https://github.com/sympy/sympy/issues/14141)